EXECUTIVE 8-PAGE PRESENTATION DECK

AeroBook Cloud Airline Platform

Technical Architecture, Microservices Topology, UML Models, Relational Persistence, Business Execution, Challenges Overcome & Production Roadmap.

SLIDE 01
System Architecture
Netty Ingress, Eureka Discovery, OpenFeign & AMQP Bus.

SLIDE 02
Use-Case Model
Passenger, Staff & Admin RBAC with Razorpay & SMTP.

SLIDE 03
Class Diagram
Domain entities, lifecycle enums & DDD associations.

SLIDE 04
Relational ER Model
8 isolated MySQL schemas with explicit PK/FK/UK constraints.

SLIDE 05
Business Activity
Search, Coupon, Payment Fork/Join & Gate Verification.

SLIDE 06
Data Flow (DFD)
End-to-end packet journey from client to MySQL InnoDB.

SLIDE 07
Challenges Overcome
Distributed consistency, race conditions & auth throughput.

FUTURE SCOPE
Production Roadmap
Redis Caching, CircuitBreakers, Flyway & WebSockets.

Platform: Java 21 LTS • Spring Boot 3.3.4 • Angular 18 • MySQL 8 • RabbitMQ

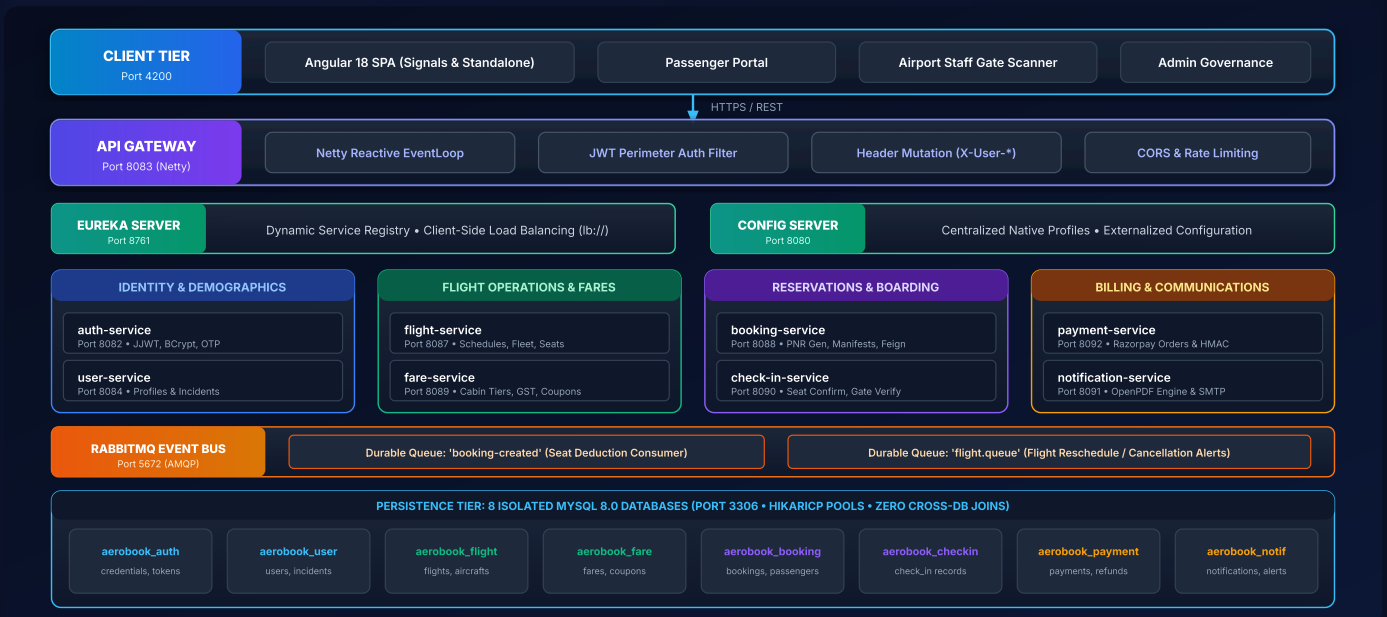
Author: AeroBook Platform Engineering & Architecture Team

October 2026 • 8-Page Executive Presentation

AeroBook

System Architecture & Microservices Topology

Multi-Tier Enterprise Cloud Architecture: Netty Ingress, Eureka Discovery, AMQP Event Bus & Persistence



SPEAKER TALKING POINTS

- Single Point of Ingress: Client interacts exclusively through Netty API Gateway (8083), hiding all internal service ports.
- Perimeter Security: Cryptographic JWT token validation occurs at the Gateway, injecting X-User-Id and X-User-Role claims downstream.
- Hybrid Communication: OpenFeign for low-latency synchronous RPC queries + RabbitMQ for decoupled, reliable event delivery.

AeroBook

System Use-Case & RBAC Governance Model

Actor Role Boundaries, Operational Capabilities, Permission Matrices & Third-Party Integrations



SPEAKER TALKING POINTS

- Role Partitioning: Strict boundaries isolate Passengers (ROLE_USER), Airport Staff (ROLE_STAFF), and Admins (ROLE_ADMIN).
- Departure Gate Integration: Staff operates digital gate scanner (/verify) that directly transitions passenger status to BOARDED.
- External Collaborators: Razorpay handles order payments; Gmail SMTP dispatches vector PDF boarding passes and OTPs.

AeroBook

Domain Class Diagram & Modular DDD Architecture

Domain Entities, Value Objects, Structural Relationships & Typed Lifecycle Enumerations



SPEAKER TALKING POINTS

- Domain Separation: Authentication credentials (Credential) are decoupled from user demographics (UserEntity).
- Immutable Audit Keys: PNR (Passenger Name Record), boarding pass codes (BP-*), and Razorpay order IDs are unique across services.
- Lifecycle Enums: State machines are strictly governed via typed enums (FlightStatus, BookingStatus, PaymentStatus, IncidentStatus).

AeroBook

Entity-Relationship (ER) Persistence Model

Relational Schemas, Constraints, Keys & Foreign References across 8 Isolated MySQL Databases



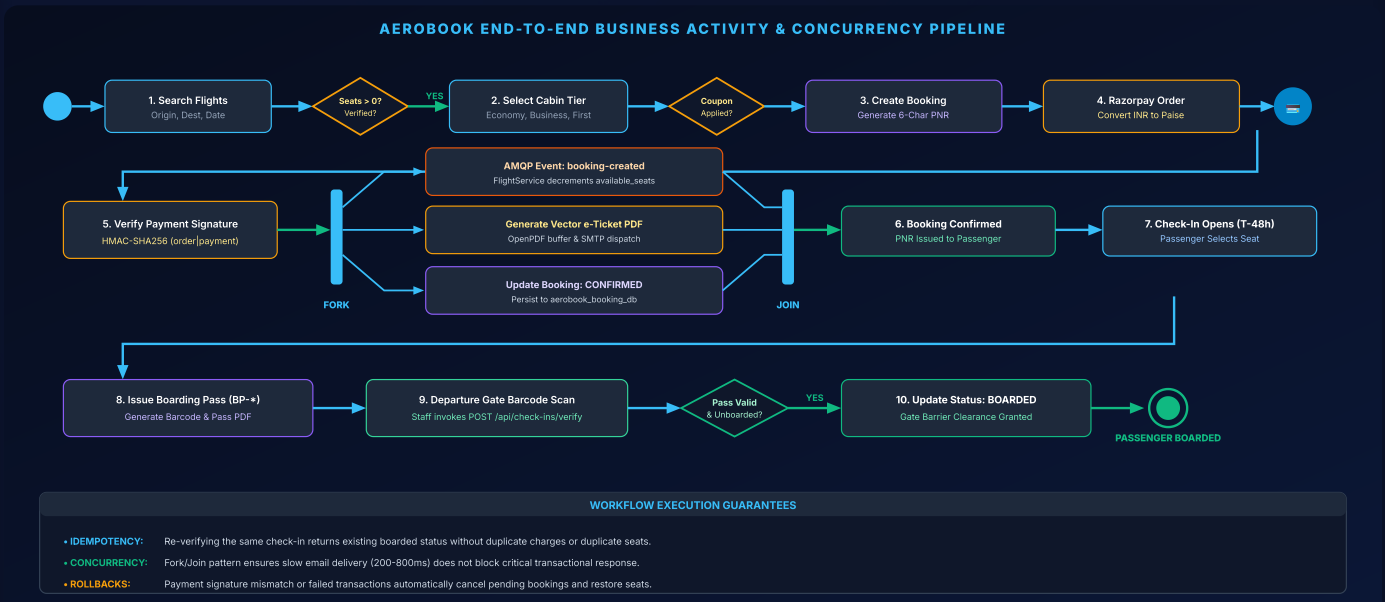
SPEAKER TALKING POINTS

- Database-per-Service: 8 private MySQL schemas prevent cross-service deadlocks and allow autonomous schema scaling.
- Intra-Domain Cascades: Direct foreign keys enforce relational integrity within services (e.g. bookings to passengers).
- Master Seed Dataset: 20 user credentials, 15 aircraft, 30 scheduled commercial flights, and active promotional coupons.

AeroBook

End-to-End Business Activity & Lifecycle Workflow

Procedural Business Execution: Flight Search, Coupon Validation, Razorpay Payment, AMQP Sync & Gate Clearance



WORKFLOW EXECUTION GUARANTEES

- **IDEMPOTENCY:** Re-verifying the same check-in returns existing boarded status without duplicate charges or duplicate seats.
- **CONCURRENCY:** Fork/Join pattern ensures slow email delivery (200-800ms) does not block critical transactional response.
- **ROLLBACKS:** Payment signature mismatch or failed transactions automatically cancel pending bookings and restore seats.

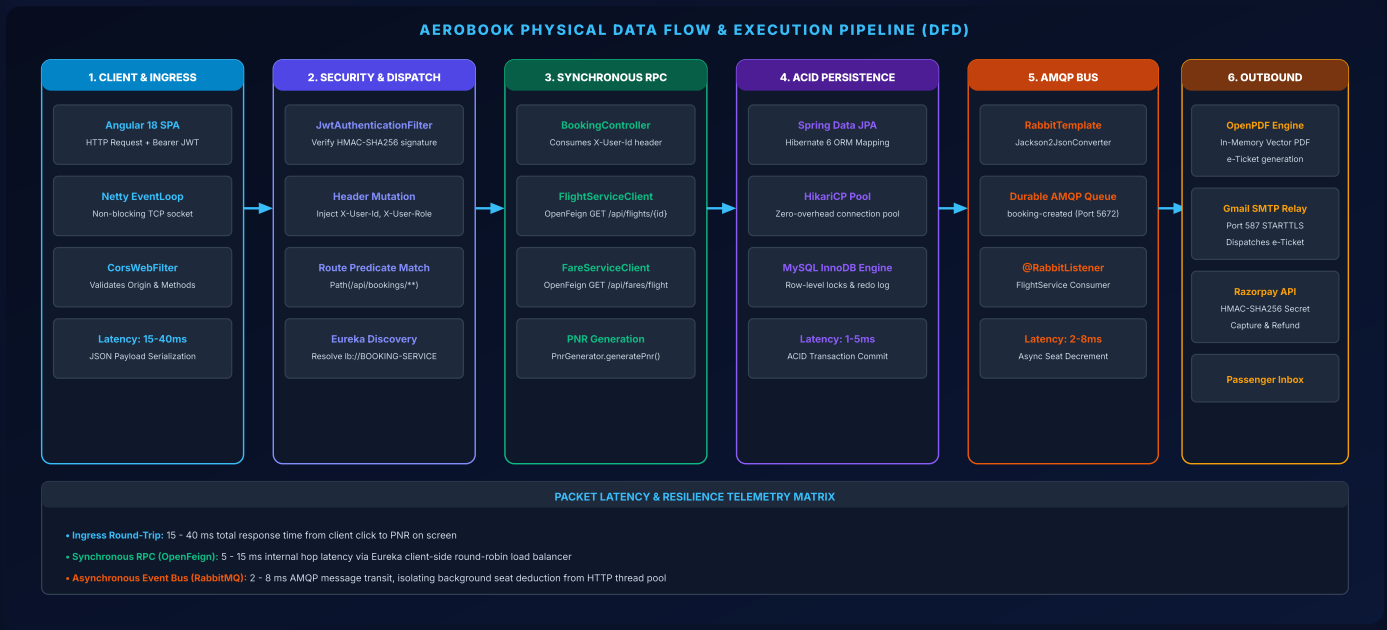
SPEAKER TALKING POINTS

- Validation Checkpoints: Seat availability, coupon discount eligibility, and HMAC-SHA256 signature checking.
- Fork/Join Concurrency: Successful payment concurrently publishes AMQP seat reduction, renders e-ticket PDF, and confirms booking.
- Gate Clearance Handshake: Airport staff mobile barcode scan transitions passenger records to BOARDED at departure gate.

AeroBook

System Execution & Distributed Data Flow Pipeline (DFD)

Physical Packet Traversal: Netty Ingress, JWT Mutation, Eureka Dynamic RPC, AMQP Bus & DB Transactions



SPEAKER TALKING POINTS

- Packet Ingress: HTTP request enters Netty EventLoop on port 8083, passes CorsWebFilter and JwtAuthenticationFilter.
- Dynamic Routing: Spring Cloud Gateway queries Eureka cache to resolve lb://BOOKING-SERVICE via round-robin load balancing.
- Decoupled Pipelines: Synchronous OpenFeign for seat checks (5-15ms); asynchronous RabbitMQ for seat deduction (2-8ms).

AeroBook

Technical Challenges Overcome & Future Roadmap

Distributed Data Consistency, Perimeter Security, Concurrency, Redis Caching & Resilience4j Circuit Breakers



SPEAKER TALKING POINTS

- Highlight Distributed Consistency solution: OpenFeign for read validation + RabbitMQ for eventual consistency.
- Explain Race Condition resolution: Pessimistic locking (SELECT FOR UPDATE) in Flight Service prevents flight overselling.
- Outline Future Scope: Redis cluster caching, Resilience4j circuit breakers, Flyway migrations, and WebSockets/SSE.